MICROBIOLOGY

UNIT 3 NOTES

FUNGI AND VIRUSES

- MORPHOLOGY, CLASSIFICATION
- REPLICATION AND CULTIVATION

DISINFECTANTS

- CLASSIFICATION
- MODE OF ACTION
- FACTORS INFLUENCING DISINFECTION
- EVALUATION OF DISINFECTION
- STERILITY TESTING OF PRODUCTS

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FUNGI

- Fungi are eukaryotic microorganism that includes microbes such as Yeasts, Moulds, Mushrooms etc.
- The cell wall of fungi is mainly composed of chitins, glucans etc.
- They are heterotrophic in nature, means they obtain nutrients by absorbing organic materials.
- They can be either unicellular (Yeasts) or multicellular (moulds). Fungi reproduce both sexually & asexually through spores.
- Study of fungi is generally known as Mycology.

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MORPHOLOGICAL CLASSIFICATION OF FUNGI

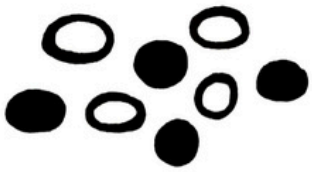
According to Morphology, fungi are mainly classified into 4 types:

- Yeast
- Yeast like Fungi
- Molds
- Dimorphic Fungi

YEAST

- They are unicellular in nature.
- They are round to oval shaped.
- They generally reproduce asexually by budding.
- They generally grow at 37°C.
- Example: Cryptococcus Neoformans

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YEAST

② MOLDS

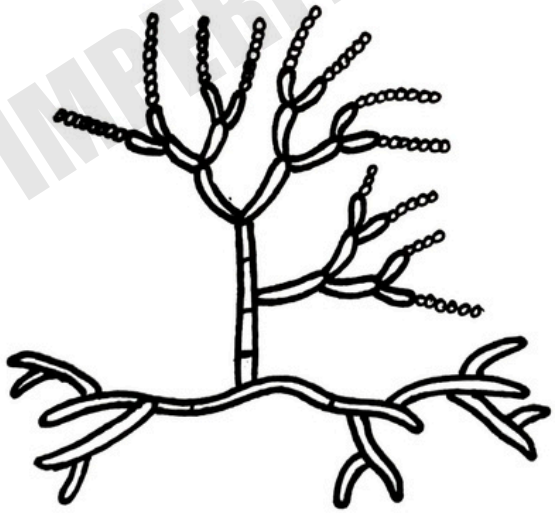
- They are multicellular in nature.
- They composed of filamentous structure called Hyphae.
- They grow at 25°C.
- They reproduce both sexually & asexually.
- Based on Hyphae, they are of two types :

ⓐ Septate Hyphae

ⓑ Aseptate Hyphae

- Examples : Rhizopus

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MOLD

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③ YEAST LIKE FUNGI

- They are similar to Yeast but have pseudohyphae
- They are unicellular in nature.
- They grow at 37°C.
- They are oval to round shaped.
- Example : Candida Albicans.





④ DIMORPHIC FUNGI

- These are those fungi that can switch between Yeast and Mold forms depending on environmental conditions :

ⓐ Yeast form (Typically at 37°C, Body Temperature)

ⓑ Mold Form (Typically at 25°C, room temperature)

- They are mostly pathogenic (disease causing) in nature.
- Example : Histoplasma capsulatum
  Blastomyces dermatitidis

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REPRODUCTION IN FUNGI

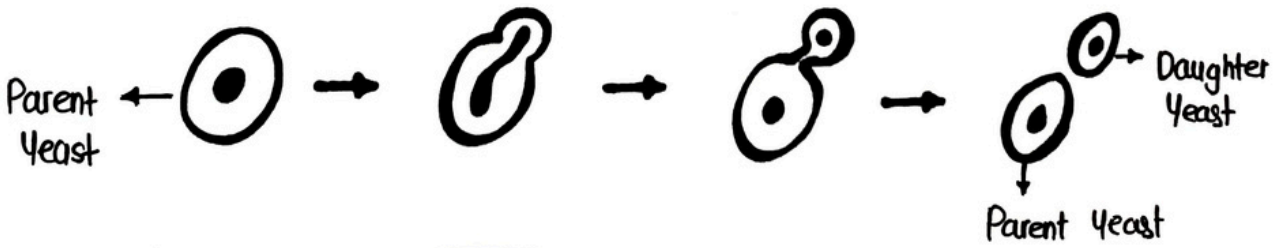
- Fungi Reproduce through both sexual & asexual methods:
- The type of reproduction depends on the species and environmental conditions.
- Reproduction in Fungi is a complex process that mainly involves production of spores, which are similar to seeds of plants.

① ASEXUAL PRODUCTION

- Asexual reproduction in Fungi is the most common form, occuring without the fusion of gametes.
- It leads to the formation of genetically identical offspring.
- There are following methods of asexual reproduction:

ⓐ Budding

- In this a small outgrowth (bud) forms on the parent cell, which detaches to become a new individual.
- Example: Saccharomyces Cerevisiae (Yeast)



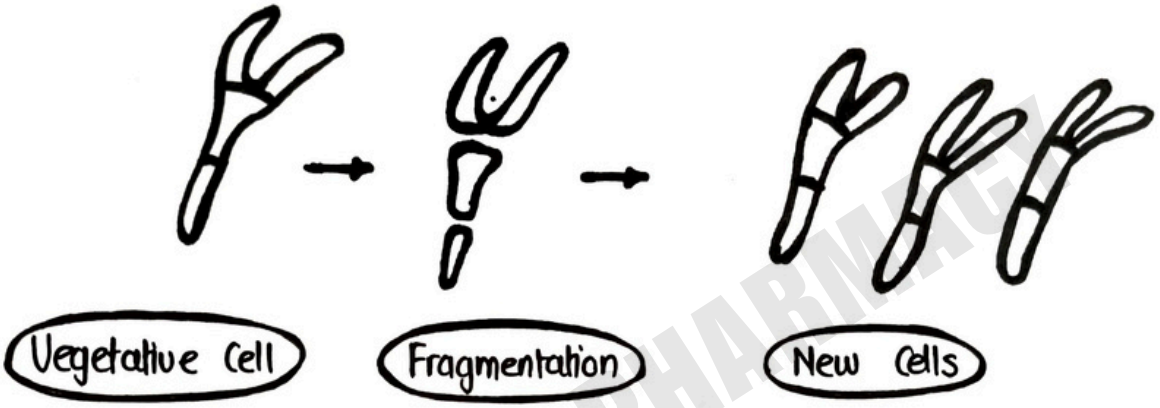
BUDDING

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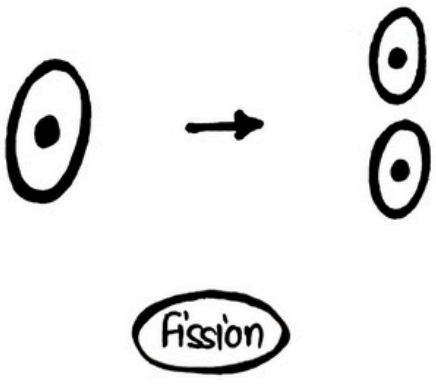
ⓑ Fragmentation

- In this hyphae of the parent fungus break into fragments, each of which grows into a new individual.
- Example: Rhizopus



ⓒ Fission

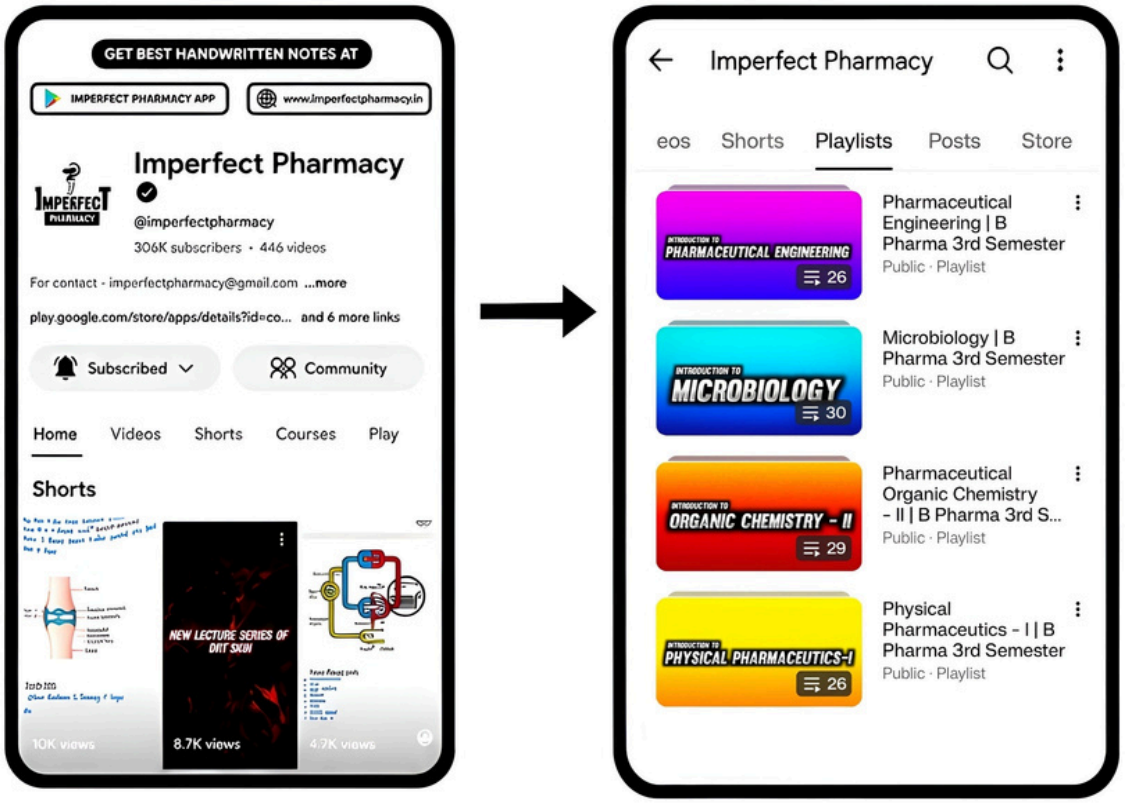
- In this parent cell divides into two equal cells.
- Example: Some yeast species like Schizosaccharomyces.



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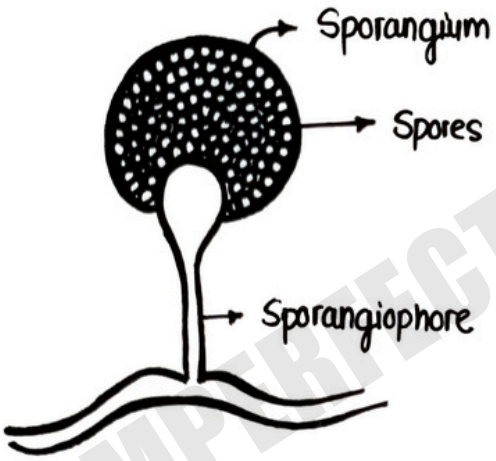
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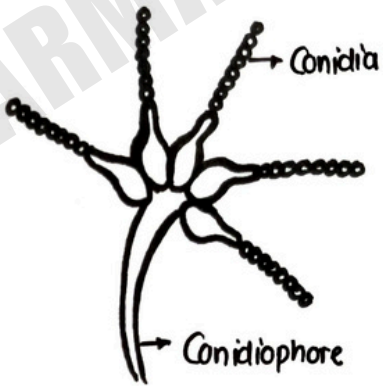
(d) Spore formation

- In this fungi reproduces through spore formation
- Some examples include

(1) Sporangiospores
(2) Conidiospores
(3) Chlamydospores
(4) Arthrospores
(5) Blastospores



Sporangiospores



Conidiospores

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② Sexual Reproduction

- Sexual Reproduction in Fungi is a process that involves the fusion of two compatible haploid nuclei, resulting in the formation of genetically diverse offspring.
- This process is crucial for maintaining genetic variation and adapting to changing environmental conditions.

Stages Of Sexual Reproduction

There are mainly 3 stages of sexual reproduction in Fungi:

① Plasmogamy
② Karyogamy
③ Meiosis

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Plasmogamy

- It is the fusion of the cytoplasm from two genetically different mating types (+ & -).
- Results in a dikaryotic stage, where two nuclei coexist in the same cell without fusing.

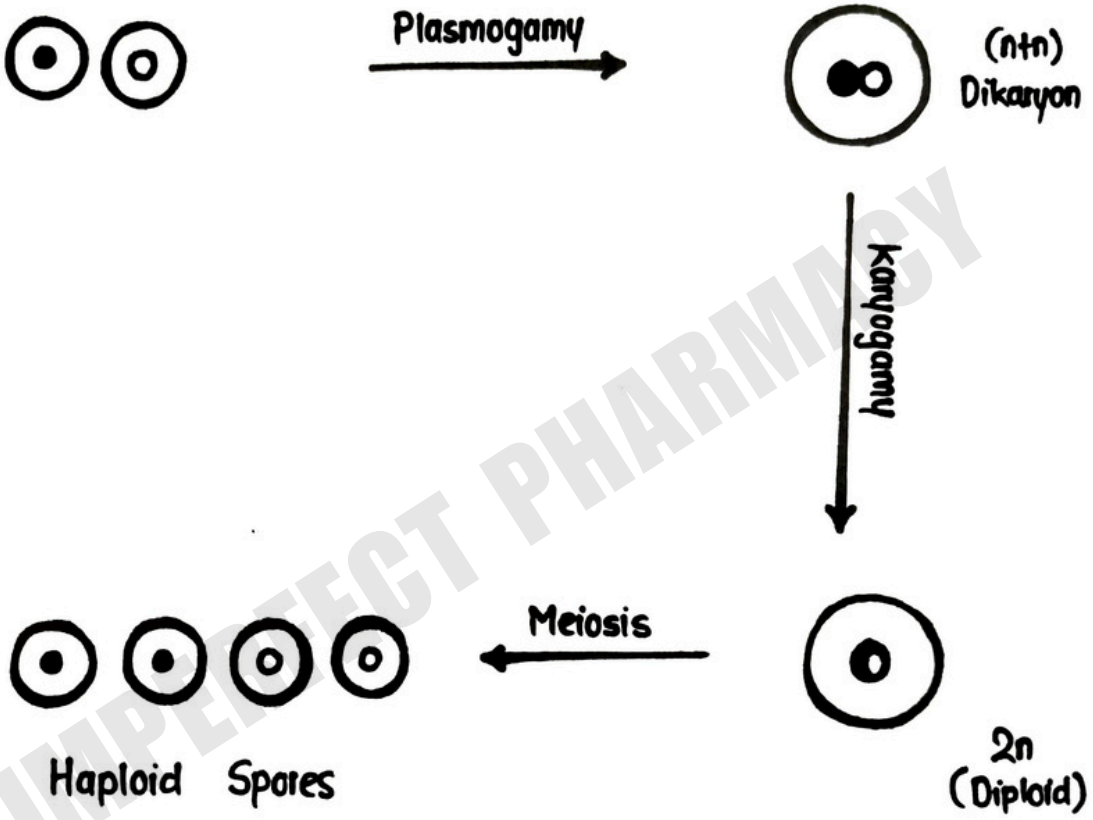
karyogamy

- In this two haploid nuclei fuse to form a diploid (2n) nucleus.
- This stage may be delayed leading to prolonged dikaryotic phase in some fungi.

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Meiosis

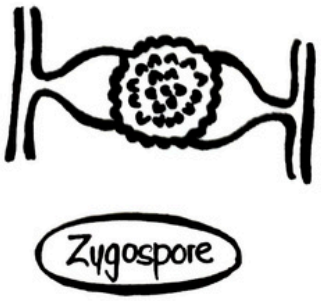
The diploid nucleus undergoes meiosis, producing haploid spores, which can germinate into new fungal organisms.



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Examples Of Sexual Spores

- Zygospores
- Ascospores
- Basidiospores
- Oospores



Zygospore

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CULTIVATION OF FUNGI

Cultivation of Fungi refers to the process of growing Fungi in a controlled environment, such as laboratory, greenhouse or industrial settings for various purposes such as :

- Food Production
- Medicinal Application
- Industrial Application
- Research etc.

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CULTURE MEDIA

- The culture medium mainly used for cultivation of Fungi is Sabouraud Dextrose Agar (SDA).
- It contains :
  - Dextrose : 4 g
  - Peptone : 1 g
  - Agar : 2 g
  - H_2O : 100 ml
- The pH of the medium is adjusted to 5.6 to enhance growth of Fungi.

PROCESS

- ISOLATION : Obtaining a fungal isolate from natural or laboratory sources.
- INOCULATION : Transferring fungal isolate to growth medium.
- INCUBATION : Providing optimal conditions for fungal growth.
- MAINTAINANCE : Subculturing & Monitoring.
- SCALE UP : Large scale production.

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VIRUSES

- Viruses are microscopic infectious agents that can only replicate inside the living cells of an organism.
- They are not living organisms rather they are considered obligate intracellular parasites, means they rely on a host cell's machinery to reproduce
- They contain either DNA or RNA as their genetic material, but never both.
- They typically range from 20 nm to 300 nm, much smaller than bacteria.
- They can infect all forms of life, including animals, plants and microorganisms.

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STRUCTURE OF A VIRUS

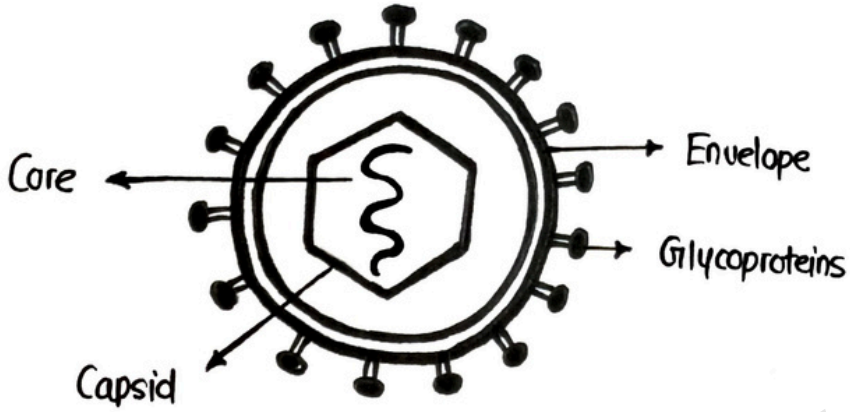
A fully assembled infectious virus is known as virion & consist of:

① Core: It is the genetic material of the virus & can be either DNA or RNA, but not both.

② Capsid: It is the protein shell surrounding the genetic material.
- It can be either icosahedral, helical or complex shaped.

③ Envelope: Some viruses have an additional lipid envelope surrounding the capsid, derived from host cell membrane.
- This envelope contains viral glycoproteins embedded on it.

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VIRUS STRUCTURE

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CLASSIFICATION OF VIRUS

Virus can be classified on the basis of two types :

- Based on Host
- Based on Morphology

① BASED ON HOST

Based on host, virus can be classified into following 4 types

ⓐ Animal Viruses
ⓑ Plant Viruses
ⓒ Bacteriophages

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@ Animal Viruses

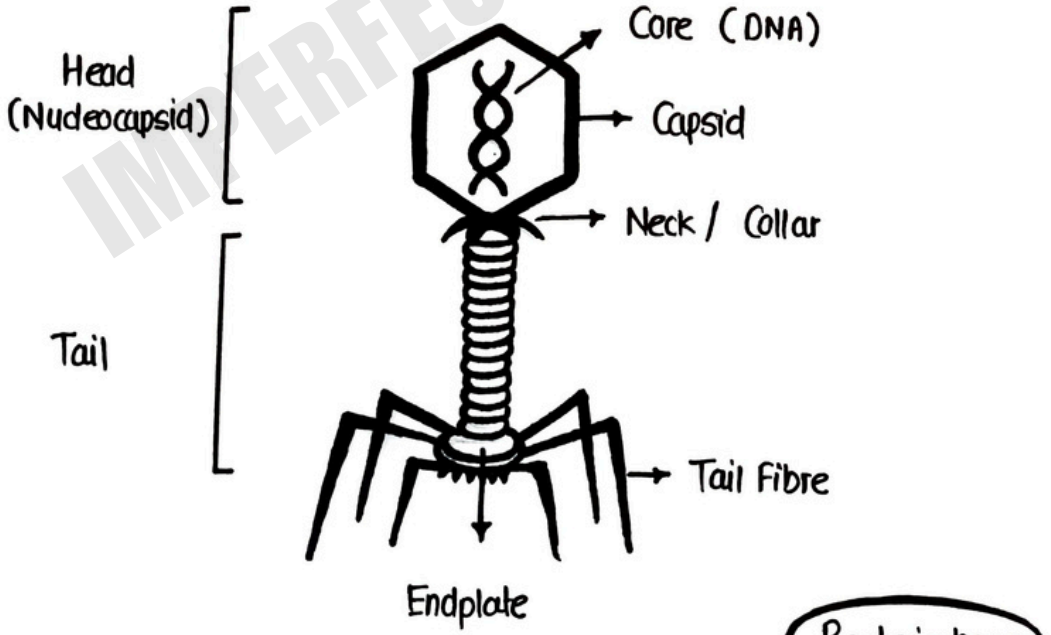
- They infect animals and humans.
- Example : Rabies virus , HIV, Influenza virus .

ⓑ Plant Viruses

- They usually have RNA & infect plants.
- Example: Tobacco mosaic virus (TMV).

© Bacteriophages

- They usually have DNA & infect bacteria
- They consist of two parts : Head & Tail.
- Example: T4 bacteriophage.



Bacteriophage

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② Based On Morphology

On the basis of morphology, virus can be classified into following types :

- Icosahedral Viruses
- Helical Viruses
- Complex Viruses
- Enveloped Viruses
- Non- Enveloped Viruses

ⓐ Icosahedral Viruses

- They are characterized by their highly symmetrical, spherical shape.
- Example : Adenovirus, Poliovirus.

ⓑ Helical Viruses

- They are characterized by their rod like or filamentous shape.
- Example : Tobacco mosaic virus, Rabies virus.

ⓒ Complex Viruses

- They have complex structure with additional features, such as tail fibres.
- Example : T4 bacteriophage.

ⓓ Enveloped Viruses

- They are surrounded by a lipid membrane derived from host cell.
- Example : HIV, Influenza virus.

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⑥ Non- Enveloped Viruses

- They lack an envelope, also known as naked viruses.
- Example: Norovirus

VIRAL REPLICATION

The viral replication also known as life cycle of virus is complex process that involves several stages:

① Attachment

- In this, virus binds to specific receptors on host cell using proteins or glycoproteins on its surface.

② Entry

- The virus or its genetic material enters the host cell.
- This can happen through direct fusion or endocytosis.

③ Uncoating & Replication

- After entry, the viral capsid is removed, exposing the viral genome.
- Viral genome starts replicating, using host cell machinery.

④ Biosynthesis

- Viral proteins are synthesized using host ribosomes.

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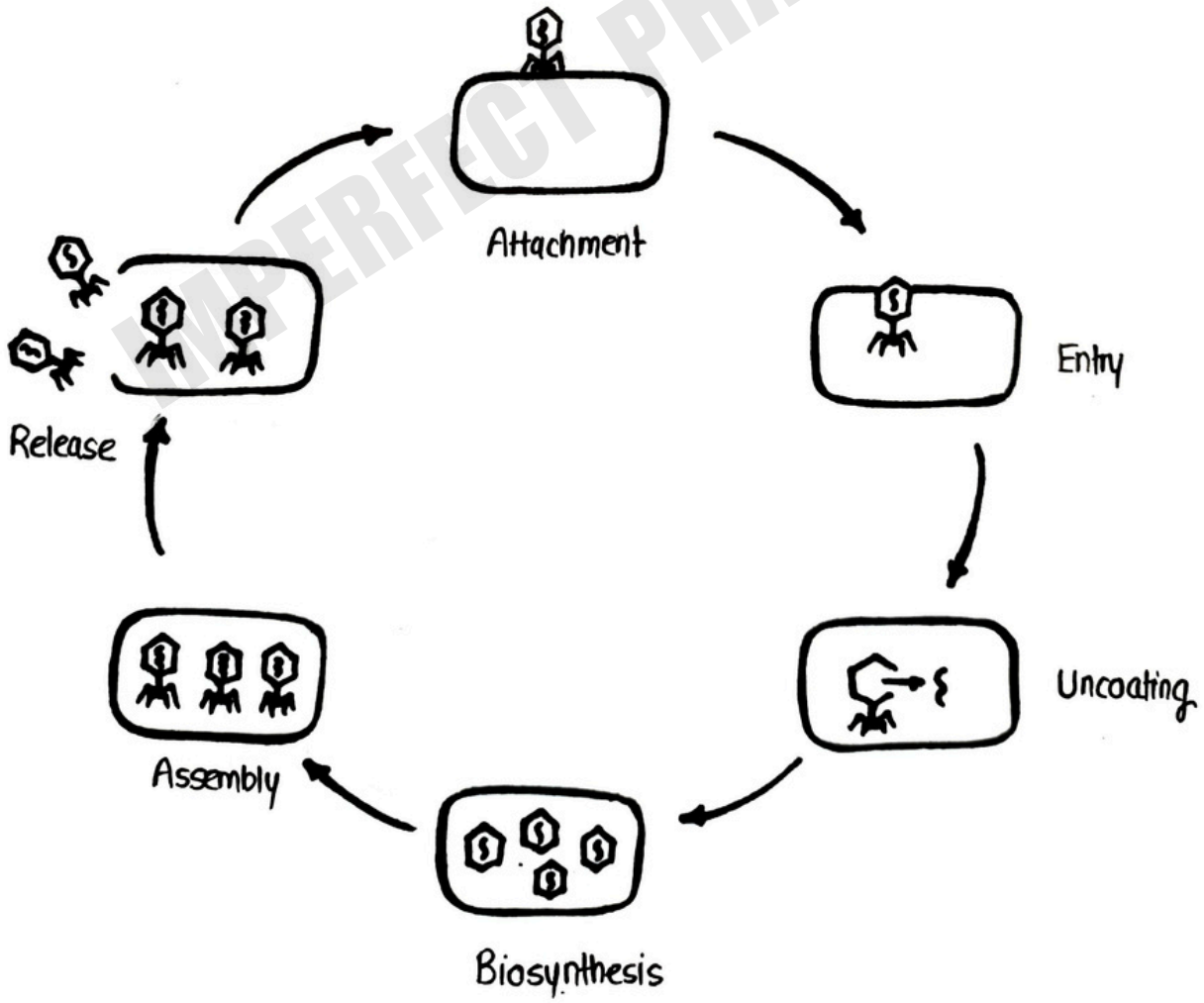
⑤ Assembly / Maturation

- Newly synthesized viral genomes and proteins are assembled into new viral particles (virions)

⑥ Release

- The newly synthesized viruses exit the host cell to infect other cells by two ways.

(a) Budding : Enveloped Viruses
(b) Cell lysis : Non- Enveloped Viruses



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CULTIVATION OF VIRUSES

- Viruses are obligate intracellular parasites, means they require living host cells to grow and replicate.
- Cultivating viruses involves providing suitable host systems such as living organisms, tissues or cell cultures.
- This process is essential for studying viral properties, producing vaccines and diagnosing viral diseases.

Methods Of Viral Cultivation

Viruses are cultivated by following methods :

- Cultivation in Animal
- Cultivation in Embryonated Egg
- Cultivation in Cell Culture

① CULTIVATION IN ANIMAL

- In this viruses are injected into specific organisms or tissues of animals like mice, rabbits or guinea pigs etc.
- Example: Rabies virus is grown in mice.
- It is mainly used for pathogenic studies & vaccine development.

② CULTIVATION USING CELL CULTURE

- It involves growing viruses in artificial environments like cell cultures derived from tissues.
- Example: Monkey kidney cells for polio virus.
- They are short lived & difficult to maintain.

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③ Cultivation In Embryonated Eggs

- The cultivation of viruses in embryonated eggs is a widely used method for isolating, growing and studying viruses, especially for vaccine production.
- It involves using a fertilized chicken egg (typically around 7 - 10 days old)
- Viruses are inoculated into different parts of the egg, depending on virus type :

(1) Chorioallantoic Membrane (CAM) : Herpesvirus
(2) Amniotic Sac : Influenza Virus
(3) Allantoic Cavity : Newcastle disease virus
(4) Yolk Sac : Yellow fever Virus

Advantages

- Cost effective & easily available.
- Suitable for large scale vaccine production.

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DISINFECTION

- Disinfection is the process of eliminating or reducing harmful microorganisms such as bacteria, viruses, fungi from surfaces objects or living beings.
- Disinfection does not necessary kill all microorganisms, but it reduces their number to a level that is considered safe based on public health standards.
- Disinfection is mainly performed using these two chemical agents:

① Disinfectant
② Antiseptic

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① DISINFECTANTS

- Disinfectants are chemicals used to kill or reduce harmful microorganisms on inanimate (lifeless) objects or surfaces.
- Disinfectants tends to be stronger & can be too harsh for use on living tissues.
- Example: Bleach, Hydrogen peroxide, alcohol based cleaners etc.

② ANTISEPTICS

- Antiseptics are chemicals used to prevent or stop the growth of microorganisms on living tissues, such as skin or mucous membrane.
- They are less harsh than disinfectants to avoid damaging living tissue.
- Example: Hand sanitizers, Iodine solutions etc.

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Ideal Characteristics Of Disinfectants

- It should have a broad spectrum of activity.
- It should be fast acting.
- It should be non-toxic.
- It should be active at all pH.
- It should be stable.
- It should have a long shelf life.
- It should have a high penetration power.
- It should be inexpensive and easily available.
- It should have a pleasant odour.

Classification & Mode Of Action Of Disinfectants

Different type of disinfectants shows different mode of action.
Here is classification of disinfectants along with their moa.

- Phenols & Phenolic compounds
- Alcohols
- Aldehydes
- Halogens
- Heavy Metals
- Oxidising Agents
- Dyes
- Detergents

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① PHENOLS & PHENOLIC COMPOUNDS

- Mode of Action : They act by disrupting the cell membrane, precipitating the proteins and inactivating microbial enzymes.
- Examples : Phenol, Cresol, triclosan etc.
- Use : commonly used in hospitals and homes.

② ALCOHOLS

- Mode of Action : Denature proteins and disrupt cell membranes of microorganisms.
- Examples : Ethanol, Isopropanol.
- Use : They are widely used as hand sanitizers and surface disinfectants.

③ ALDEHYDES

- Mode of Action : Cross-link proteins and inactivate nucleic acids.
- Examples : Formaldehyde, Glutaraldehyde
- Use : For high level disinfection and sterilization.

④ HALOGENS

- Mode of Action : They act by oxidizing cellular components of microorganisms.
- Examples : Chlorine, Iodine
- Use : They are used in water treatment & wound care.

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⑤ HEAVY METALS

- Mode of Action: They binds to cellular proteins and inactivate them
- Examples: Mercuric Chloride, silver nitrate, copper sulphate etc.
- Uses: They are mainly used in water purification and as surface disinfectant

⑥ OXIDISING AGENT

- Mode of Action: Oxidize cellular components.
- Examples: Hydrogen Peroxide, peracetic acid.
- Uses: Used in medical and food processing settings.

⑦ DYES

- Mode of Action: They act by interference with cell membrane integrity and protein denaturation.
- Examples: Crystal violet, acriflavine, malachite green etc.
- Uses: They are mainly used as wound antiseptic.

⑧ DETERGANTS

- Mode of Action: They act by disrupting cell membranes.
- Examples: Quaternary Ammonium Compounds (Quats)
- Uses: commonly used as household cleaners.

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FACTORS INFLUENCING DISINFECTION

Several factors influence the effectiveness of disinfection. Understanding these factors is essential for optimizing disinfection processes and ensuring microbial control

- Concentration of disinfectants
- Temperature
- Contact Time
- pH of the environment
- Surface condition
- Characteristics of Microorganism
- Water Hardness

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① Concentration Of Disinfectants

- Concentration of the disinfectant directly influences the rate of killing microorganisms.
- However too high concentration can lead to decrease in effectiveness.

② Temperature

- Most disinfectants are more effective at higher temperatures.
- However, extremely high temperatures can cause evaporation or degradation of some disinfectants.

③ Contact Time

- Disinfectants required adequate time of contact to kill or inhibit microorganisms.

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④ pH Of The Environment

- Some disinfectants are more effective in acidic or alkaline environments.

⑤ Surface Condition

- Rough or damaged surfaces can harbor microorganisms and make disinfection less effective.

⑥ Characteristics Of Microorganisms

- The efficacy of disinfection greatly depends on nature and number of contaminating microorganisms.
- Bacterial spores and Non-enveloped viruses are generally more resistant.

⑦ Water Hardness

- Hard water can reduce the effectiveness of some disinfectants, such as quaternary ammonium compounds.

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EVALUATION OF DISINFECTANT

Evaluation of disinfectant assess their ability to kill or inhibit the growth of microorganisms. It involves following methods:

- Tube dilution and agar plate method
- Cup-Plate Method
- Ditch-Plate Method
- Phenol coefficient Method
- kelsey Sykes Method

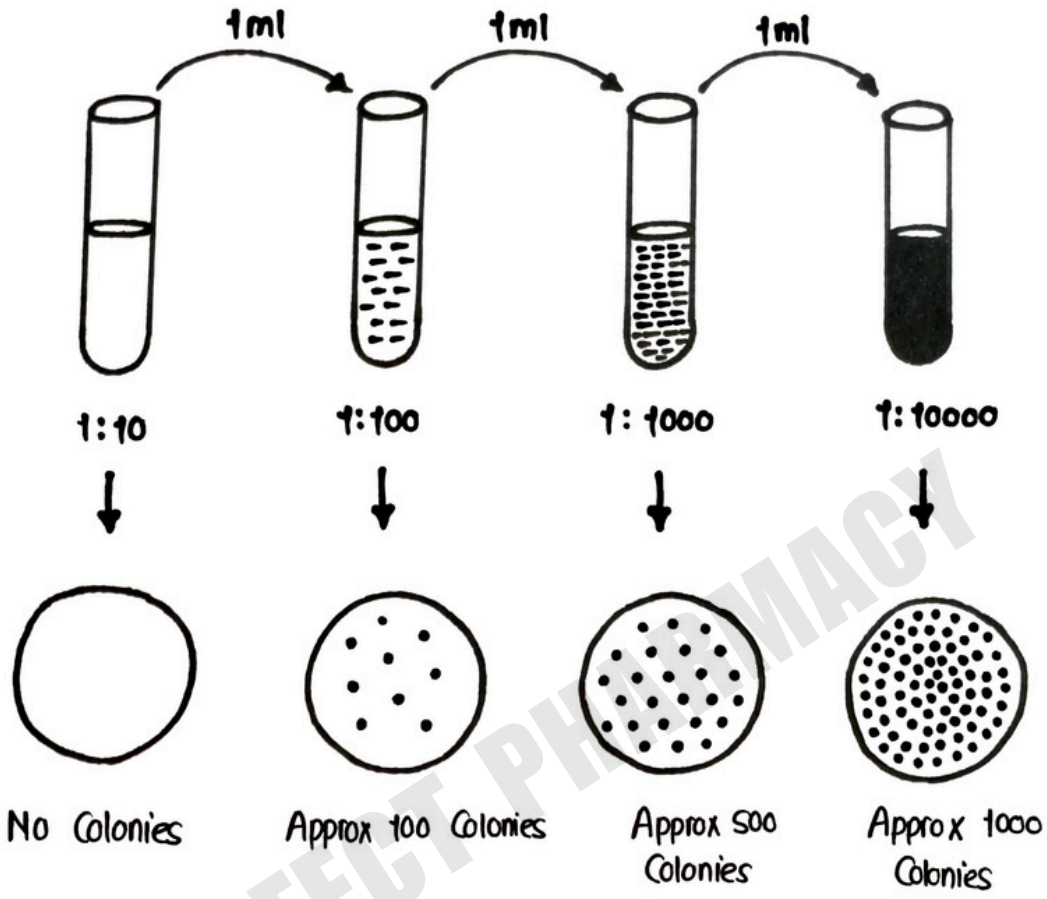
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① TUBE DILUTION & AGAR PLATE METHOD

- Tube dilution and agar plate method is a laboratory technique used to evaluate the effectiveness of disinfectants against microorganisms
- It involves following steps:

* Prepare a series of dilutions of disinfectant typically ranging from 1:10 to 1:1000.
* Inoculate each dilution with a known number of microorganism.
* Now incubate the test tubes at a specific temperature (around 37°C) for specific period of time, typically around 24-48 hour.
* Observe the test tubes or agar plates for visible growth & turbidity and record the results.

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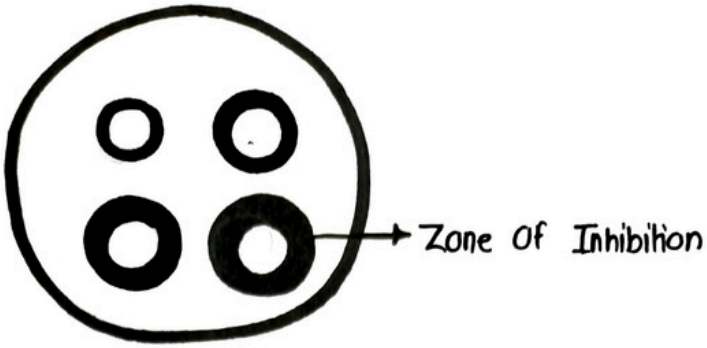
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② Cup Plate Method

- The cup plate method, also known as cylinder plate method, is a technique used to evaluate effectiveness of disinfectants
- It involves following steps:

* Pour agar medium into petri dishes and allow to solidify.
* Spread the microbial suspension onto the agar surface.
* Use a sterile borer to punch small wells into agar plate
* Carefully add a specific volume of disinfectant to each well of different concentration.
* Incubate the plate at 37°C for 24-48 hour.
* Measure the zone of inhibition around each well.

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CUP PLATE METHOD

③ DITCH PLATE METHOD

- The ditch plate method is a laboratory technique used to evaluate the antimicrobial efficacy of disinfectants.
- It involves following simple steps
* Agar plates are prepared with microorganisms streaked on it.
* A ditch or groove is made in the agar.
* Disinfectant solution is placed in the ditch.
* Measure the zone of inhibition around the ditch



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④ PHENOL COEFFICIENT METHOD

- The Phenol Coefficient method evaluates disinfectant efficacy by comparing it to phenol, a standard disinfectant
- The phenol coefficient is a number that represents the relative effectiveness of disinfectant compared to phenol.
- It involves two major test :

ⓐ Rideal - Walker Test
ⓑ Chick - Martin Test

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ⓐ Rideal Walker Test

Rideal - Walker test is a historical method used to evaluate the effectiveness of disinfectant by comparing it to phenol, which is used as a standard.
It involves following steps:

- Dilutions of both phenol & the disinfectant to be tested are prepared in sterile distilled water, the dilutions are typically made in a series, with each dilution being a fraction of previous one.
- A specific volume of standardized suspension of Salmonella typhi bacteria is added to each dilution of disinfectant and phenol.
- The tubes containing the dilutions and bacteria are incubated at a specific temperature
- After the incubation period, samples from each dilution are transferred to tubes containing sterile nutrient broth.
- These subcultures are then incubated for a further period to determine if any bacterial growth has occurred.

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DISINFECTANT	DILUTION	TIME INTERVAL			
		2·5	5	7·5	10
Test Disinfectant	1: 1000	+	−	−	−
	1: 2000	+	+	−	−
	1: 3000	+	+	+	−
	1·4000	+	+	+	+
Phenol	·1:80	+	−	−	−
	1:100	+	+	−	−
	1:120	+	+	+	−
	1:140	+	+	+	+

$$\text{R·W· Coefficient} = \frac{\text{Dilution of disinfectant killing in 7·5 min. but not in 5m.}}{\text{Dilution of phenol killing in 7·5 min. but not in 5 min.}}$$

$$= \frac{2000}{100}$$

$$= 20$$

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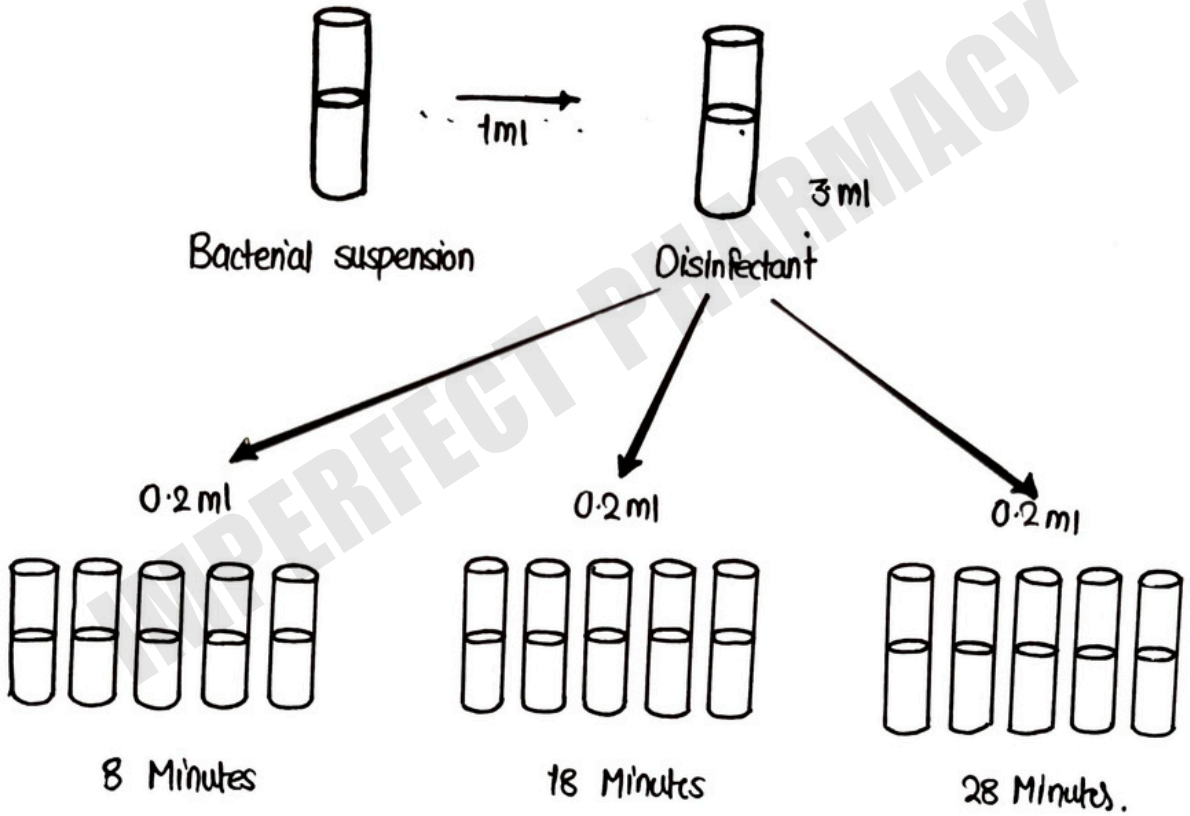
Conclusion

- If the rideal - walker test for given disinfectant is 1, the disinfectant has same effectiveness as phenol.
- If RW· coefficient is 20, then disinfectant is 20 times more active than phenol

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⑤ KELSEY-SYKES TEST

- The kelsey-sykes test is a method used to evaluate the bactericidal effectiveness of disinfectant.
- It involves mixing the disinfectant with bacterial suspension and then subjected to three successive challenges.
- The number of surviving bacteria is counted to assess the disinfectant's efficacy.



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STERILITY TESTING

- Sterility Testing is defined as a testing which confirms that products are free from presence of microorganisms.
- It is a critical quality control process used in pharmaceutical, microbiology, biotechnology and medical device industries to ensure that product are free from viable microorganisms.

PRINCIPLE OF STERILITY TESTING

- The principle of sterility testing is based on the concept of providing optimal conditions for growth of microorganisms
- If a product is contaminated with viable microorganism, they will multiply and grow and their presence can be indicated by turbidity in the original ~~colour~~ clear medium

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CULTURE MEDIA USED FOR STERILITY TESTING

Following culture medias are mainly used for Sterility Testing

1. Fluid Thioglycolate Medium
2. Alternative Thioglycolate Medium
3. Soyabean Casein Digest Medium

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① Fluid Thioglycolate Medium

It is mainly used for clear fluid products and suitable for growth of anaerobic bacteria. It composed of :

- L- Cystine
- Sodium Chloride
- Dextrose Monohydrate
- Yeast Extract
- Pancreatic digest of casein
- Sodium Thioglycolate
- Dipotassium hydrogen Phosphate
- Resazurin
- Distilled Water

② Alternative Thioglycolate Medium

This medium is mainly used for turbit and viscid medium and mainly consist of :

- L- Cystine
- Sodium Chloride
- Dextrose Monohydrate
- Yeast Extract
- Pancreatic digest of casein
- Sodium thioglycollate
- Distilled Water

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Soyabean Casein Digest Medium

It is suitable for culturing both Fungi and anaerobic bacteria and mainly consist of :

- Pancreatic digest of casein
- Papaic digest of soyabean meal
- Sodium chloride
- Dibasic potassium phosphate
- Dextrose monohydrate
- Distilled water

MEDIA SUITABILITY TEST

The test is perform prior to sterility testing for culture media to make sure that :

- Media is sterile
- Media supports growth of microorganisms

It involves two test

① Media Sterility Test
② Growth Promotion Test

① Media Sterility Test

- Media sterility test is a quality control test used to verify that microbial growth medium is sterile and free from contaminants.
- In this medium is incubate for 14 days.
- The medium should show no visible sign of microbial growth.

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② Growth Promotion Test

- A growth promotion test is a quality control test used to verify that a microbiological culture medium is capable of supporting the growth of microorganisms.
- The test involves inoculating the medium with a small amount of standardized microbial suspension and then incubating it for a specific period.
- The medium should show visible sign of microbial growth.

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METHODS FOR STERILITY TESTING

Sterility Testing is performed using following 2 methods:

- Membrane Filtration
- Direct Inoculation

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① MEMBRANE FILTRATION

Membrane filtration method is a widely used technique for sterility testing of products, especially in microbiology.

Principle

The method relies on filtering a liquid sample through a membrane filter that retains microorganisms, allowing for their subsequent growth in suitable culture medium.



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Procedure

- Sample Collection
- Filtration setup using a filtration apparatus equipped with a sterile membrane filter paper, typically with pore size of 0.45 µm.
- Pour the sample onto membrane filter ensuring process is conducted under aseptic conditions.
- After filtration, place the membrane filter into culture medium.
- Regularly check for microbial growth.

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Result

- No growth indicates sterile sample.
- Growth indicates presence of microorganism.

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② Direct Inoculation Method

Direct Inoculation method of sterility testing involves adding a sample directly to a sterile growth medium, such as Agar broth to detect presence of microorganisms

Principle

- Microorganism, if present in the sample will grow in nutrient rich medium.
- Sterile sample will not support microbial growth.

Procedure

- Prepare the sample
- Inoculate the sample directly into sterile growth medium
- Incubate the medium at specified temperature for required time
- Observe the microbial growth

Result

- No growth indicates sterile sample.
- Growth indicates presence of microorganisms.

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THANK YOU

FOR CHOOSING IMPERFECT PHARMACY AS YOUR STUDY PARTNER

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